Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
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Summary

Your summary goes here. It should concisely state the problem, the model framework, key assumptions, major results, and practical implications.

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1 Introduction

1.1 Overview

Post-secondary education is undergoing a fundamental transformation as generative artificial intelligence (Gen-AI) influences both pedagogical practices and career pathways. AI-enhanced analysis, decision-making, and content creation are rapidly becoming core competencies across many professional fields. Though Gen-AI literacy has gained increasing attention in academia and industry, the degree to which it should be used is put into question. In institutions, there is a limited understanding of how to integrate Gen-AI as a complementary tool to support existing curricula. Moreover, it remains unclear how Gen-AI impacts career trajectories. This research aims to model Gen-AI's stratified influence across different educational pathways — STEM, trade/vocational, and arts/humanities —and to explore how institutional policies address the need for pervasive and appropriate Gen-AI integration that fosters practical and ethical job-ready competencies.

1.2 Exploratory Analysis

The influence of Gen-AI on career prospects varies substantially across post-secondary education pathways, yet this stratification remains underexamined in current literature.

1.2.1 STEM

STEM education prepares students for technical and analytical roles where Gen-AI serves primarily as an augmentation tool—enhancing computational capabilities, accelerating code generation, automating data analysis, or assisting in technical writing. Labor market data indicates that STEM occupations show heterogeneous Gen-AI exposure: software development faces high Gen-AI penetration in code generation and debugging, while specialized engineering fields involve more limited but growing Gen-AI integration in simulation and optimization tasks [**?**]. These careers require students to develop complementary skills: understanding Gen-AI system capabilities and limitations, validating algorithmic outputs, refining prompts for technical tasks, and integrating Gen-AI tools into complex problem-solving workflows. The pedagogical challenge lies in teaching students not merely to use Gen-AI, but to critically evaluate its outputs and leverage it strategically within technical domains—skills that must be embedded across core STEM courses, not confined to standalone AI classes.

1.2.2 Trade and Vocational

Trade and vocational schools train students for skilled manual and technical occupations—electricians, medical technicians, automotive specialists, construction workers—where Gen-AI's impact manifests differently. Employment projections suggest these occupations face lower direct displacement risk but increasing integration of Gen-AI-enhanced tools [**?**]. While Gen-AI diagnostic assistants, automated scheduling systems, or predictive maintenance algorithms may enter these fields, core competencies remain grounded in hands-on expertise, equipment operation, and situational judgment that resists full automation. Gen-AI may augment documentation (generating work reports), client communication (drafting service proposals), or diagnostic processes (interpreting equipment sensor data), but the fundamental craft skills remain human-centered. These educational pathways face a dual challenge: determining which Gen-AI literacies are genuinely transferable to trades contexts, and avoiding the assumption that Gen-AI integration follows the same trajectory as in knowledge-work sectors. Curriculum adaptation here may involve teaching

students to interact with Gen-AI-enhanced diagnostic equipment or use Gen-AI for administrative tasks, rather than direct content generation.

1.2.3 Arts and Humanities

Arts and humanities programs prepare students for creative, interpretive, and communication-intensive careers where Gen-AI presents paradoxical challenges. Gen-AI can now produce visual art, written content, music compositions, graphic designs, and video content—capabilities that directly compete with entry-level creative work [**?**, **?**]. Simultaneously, Gen-AI creates demand for creative direction, prompt engineering, ethical oversight, content curation, and authenticity verification—skills that require deep humanistic training. Labor market analyses reveal divergent trends: routine creative tasks face displacement pressure, while roles requiring creative judgment, cultural interpretation, and ethical reasoning show increased demand for human–AI collaboration capabilities [**?**]. The pedagogical question becomes: how do arts and humanities educators cultivate distinctly human creative and critical capacities while acknowledging that students will likely collaborate with—or compete against—Gen-AI tools professionally? This sector faces unique tensions between preserving humanistic values and preparing students for Gen-AI-saturated creative industries. Crucially, this adaptation cannot be achieved through standalone "AI for Artists" courses alone—it requires transforming how studio art, creative writing, film production, and humanities seminars are taught.

1.3 Choosing Our Focus

We believe that post-secondary institutions and industries are currently facing the same operative question: how urgent, and to what necessity, does a course, program of study, or occupation need to accommodate Gen-AI? To tackle this question, we need to account for the relationship between the three instead of treating them as separate entities. For example, we want to know how the influence of AI on a program of study affects the employment demands of a certain occupation, or vice-versa, how does the influence of AI on employment inform course policies. There, the question becomes: through which medium should we establish such connections?

First, we must recognize that there remains a gap between the course itself and the occupation itself concerning AI use. For instance, if we're only given the information that one mathematics course relies heavily on AI, that does not say anything about the AI use in mathematics-related careers or jobs such as actuary, operations research analyst, data scientist, etc. By this realization, we consider the approach where we break downthe course and job into multiple shared components (skills/knowledge/tasks). These components will be key tenets that bridge between the course and the occupation.

It is crucial to acknowledge that AI use is viewed as a matter of urgency, rather than a binary decision, as different industries may hold different expectations of AI. The same perspective should be applied to courses. By breaking down to those three elements, we would develop more specific and tailored suggestions in weighing the importance of AI across those dimensions.

Therefore, our model aims to produce element-wise interpretability (skills/knowledge/task) in both courses and occupations in regard to AI implementation. By adopting our approach, stakeholders will be able to answer questions like: Which parts of a course should be revised given the influence of AI? What particular skills/knowledge/tasks should students improve on in order to keep up with the pace of AI in contemporary society? These are the questions that we'll delve into in later sections.

2 Methodology

2.1 Positioning

To make the problem tractable without losing relevance, we intend to position our solution to an intermediate-layer representation that sits between the education side and the employment side. Governments typically reason about labor-force structure and occupational evolution; educational institutions reason about curriculum design and student development; employers reason about task structures and performance expectations. Without a shared representational layer, these stakeholders lack a common scale for discussing "what AI is actually affecting." Our solution seeks to provide that common layer in a form that is explainable, computable, and auditable.

2.2 A Tri-fold Approach

In practice, directly simulating career planning is generally not realistic as a primary modeling objective: career pathways depend heavily on individual preferences, institutional constraints, regional labor markets, industry cycles, and other non-structural factors that are difficult to formalize. By the same logic, a school's courses and programs are also not ideal as the model's primary object, because curricular design and implementation vary widely across instructors, departments, accreditation regimes, and institutional missions. Therefore, we identify a shared, computable intermediate layer across education and employment—capability structure—and model it through three common elements: Skills, Knowledge, and Tasks (Working Activities). Our core objective is to mathematically represent these elements, estimate GenAI penetration at the element level, and then aggregate upward to quantify GenAI penetration for courses/programs (education side) and occupations (employment side). For each element, we decompose it into three dimensions and encode it as a vector in a three-dimensional space; both (i) the element's importance to a target object and (ii) GenAI's influence on that element are defined within the same dimensional space. Through vector interactions and weighted aggregation, the framework produces interpretable indices that approximate GenAI penetration for both educational and occupational targets. When evaluating a specific target, we apply a tri-fold approach to decompose the question: how urgent—and to what necessity—does a course, program of study, or occupation need to adapt to GenAI? 1. Decomposition and weighting: Represent the target object (e.g., an individual student, a course, an occupation, or an academic program) as a structured combination of Skills/Knowledge/Tasks and determine element weights using appropriate mechanisms such as surveys, expert elicitation, or observational studies. 2. Model execution: Input the element set and corresponding importance weights into the vector-based computation to generate (a) an overall target-level index and (b) element-level indices for the constituent Skills/Knowledge/Tasks. 3. Interpretation for action planning: Use the resulting indices, together with the target's current state and existing GenAI penetration, to assess whether and where additional GenAI integration is warranted (e.g., redesigning learning activities, updating assessments, or recommending capability migration for students).

3 Dataset and Variable Definitions

3.1 Data Entities and Primary Keys

O*NET is a federally sponsored occupational information system that provides a research-friendly, standardized way to represent an occupation as a structured bundle of skills, knowledge, and work activities, each with importance-style ratings—a schema that closely matches our modeling need to decompose "employment" into measurable elements before recombining them into indices. Because the O*NET content

model and taxonomy are formally documented and designed for consistent occupational description (including alignment with the U.S. Standard Occupational Classification ecosystem), it supports reproducible mapping across occupations and downstream datasets rather than ad-hoc labels. Finally, O*NET is not just a static database: it is distributed as part of a broader U.S. labor-market infrastructure that includes career exploration tools, making it a plausible "shared vocabulary layer" for educational settings where career development, advising, and program-to-occupation translation are central.

We adopt **O*NET-SOC codes** as occupation identifiers and **Element IDs** as identifiers for skills, knowledge areas, and tasks, ensuring consistency with the O*NET data dictionary and full traceability across datasets.

Table 1: Data entities and primary keys

Entity	Symbol	Primary Key	Source	Notes
Job / Occupation	$j \in \mathcal{J}$	O*NET-SOC	O*NET	Based on the O*NET occupational classification system
Skill	$s \in \mathcal{S}$	Element ID	O*NET Skills	Associated with occupation-specific importance scores
Knowledge	$k \in \mathcal{K}$	Element ID	O*NET Knowledge	Associated with occupation-specific importance scores
Task	$t \in \mathcal{T}$	Element ID	O*NET Tasks	Associated with occupation-specific importance scores
Course	$c \in \mathcal{C}$	Course code	Institution / Curriculum	Mapped to $\mathcal{S}$, $\mathcal{K}$, and $\mathcal{T}$ through curriculum alignment

3.2 Element-Specific Coordinate Systems

Although **Skills, Knowledge, and Tasks** are represented in distinct three-dimensional coordinate systems, reflecting fundamental differences in their epistemic and operational nature. Vector operations are defined within element types only, and cross-type aggregation occurs strictly at the index level.

For each element type (Skills, Knowledge, Tasks) and for each dimension within its corresponding vector space, raw scores are first converted into a relative scale using a rank-based normalization procedure. This approach emphasizes *relative structure* across elements and avoids sensitivity to absolute score magnitudes.

Step 1: Dimension-wise Ranking. Fix an element type $t \in \{\text{skills}, \text{knowledge}, \text{tasks}\}$ and a specific dimension $d \in \{1, 2, 3\}$. For all elements x of that type, we rank elements from lowest to highest according to their raw dimension-specific score:

$$\text{rank}_{x,d} \in \{1, 2, \ldots, |\mathcal{X}^{(t)}|\}$$

Step 2: Min–Max Normalization. The ranks are then mapped onto the unit interval using min–max scaling:

$$\hat{r}_{x,d} = \frac{\text{rank}_{x,d} - 1}{|\mathcal{X}^{(t)}| - 1} \qquad \hat{r}_{x,d} \in [0, 1]$$

This transformation yields a continuous, dimensionless measure of relative standing within each dimension.

Step 3: Discretization into Fixed Bins. To improve robustness and interpretability, the normalized values are discretized into a fixed number of bins. Specifically, we partition the unit interval into 20 equal-width bins in $[0, 1]$ and define:

$$r_{x,d} = \frac{1}{20} \lfloor 20\, \hat{r}_{x,d} \rfloor \qquad r_{x,d} \in \left\{0, \frac{1}{20}, \frac{2}{20}, \ldots, 1\right\}$$

Resulting Element Vector. The final raw vector for element x in space t is given by:

$$r_x^{(t)} = \left(r_{x,1}, r_{x,2}, r_{x,3}\right)$$

which is subsequently normalized (Section 3.5) to obtain the compositional vector used in alignment and index construction.

3.2.1 Skills: CTS Space

Applying the vectorization methodology above, each skill $s \in \mathcal{S}$ is represented in a three-dimensional skill-domain space $[0, 1]^3$:

- **C (Cognitive):** reasoning, problem-solving, abstraction
- **T (Technical):** tool use, procedural execution, technical proficiency
- **S (Social):** communication, coordination, interpersonal interaction

Skills are embedded in:

$$\mathbb{R}^3_{\text{CTS}}$$

3.2.2 Knowledge: NHF Space

Each knowledge element $k \in \mathcal{K}$ is represented in a three-dimensional knowledge-domain space $[0, 1]^3$:

- **N (Natural / Physical World):** natural sciences, physical systems, engineering fundamentals
- **H (Human / Social World):** social sciences, institutions, behavior, culture
- **F (Formal / Structural Systems):** mathematics, logic, formal rules, symbolic systems

Knowledge vectors lie in:

$$\mathbb{R}^3_{\text{NHF}}$$

3.2.3 Tasks: MIP Space

Each task $t \in \mathcal{T}$ is represented in a three-dimensional task-domain space $[0, 1]^3$

- **M (Material / Physical):** interaction with physical objects, environments, machinery
- **I (Informational):** processing, analyzing, generating, or transmitting information
- **P (People-Oriented):** direct interaction with, management of, or service to people

Tasks are embedded in:

$$\mathbb{R}^3_{\text{MIP}}$$

Table 2: Key variables and definitions

Variable	Symbol	Range	Description
Job-specific Importance	$\mathrm{Imp}_{j,x}$	$[0, 100]$	Relative importance of element x within occupation j
Dimension-wise rank	$\mathrm{rank}_{x,d}$	$\{1, \ldots, \lvert\mathcal{X}^{(t)}\rvert\}$	Rank of element x along dimension d within element type t
Normalized rank	$\hat{r}_{x,d}$	$[0, 1]$	Min–max normalized rank for element x along dimension d
Discretized component	$r_{x,d}$	$\{0, \frac{1}{20}, \ldots, 1\}$	Discretized vector component after binning
Element raw vector	$r_x^{(t)}$	$[0, 1]^3$	Element vector in native space $t \in \{\mathrm{CTS}, \mathrm{NHP}, \mathrm{NIP}\}$
Normalized element vector	$v_x^{(t)}$	$\sum = 1$	L1-normalized element structure vector
AI raw impact vector	$q_x^{(t)}$	$[0, 1]^3$	AI impact scores aligned to element dimensions
Normalized AI vector	$a_x^{(t)}$	$\sum = 1$	L1-normalized AI impact structure
Element alignment index	$I_x^{(t)}, \tilde{I}_x^{(t)}$	$[0, 1]$	Raw and normalized AI alignment score
In-layer weight (occupation)	$w_{j,x}^{(t)}$	$[0, 1]$	Importance-normalized weight within element type
In-layer weight (course)	$w_{c,x}^{(t)}$	$[0, 1]$	Coverage-normalized weight of element x in course c
Cross-layer weights (occupation)	$\alpha_S, \alpha_K, \alpha_T$	$[0, 1]$	Aggregation weights across element types for jobs
Sector–element weighting matrix	$\tau_{i,t}$	$[0, 1]$	Sector-specific weight on element type t for sector i
Job Index	J_j	$[0, 1]$	Occupation-level AI exposure index
Course Index	$C_c^{(i)}$	$[0, 1]$	Sector-specific course-level AI exposure index

3.3 Key Variables

3.4 Treatment of Importance

Within the model, Importance is used strictly as a relative weighting mechanism, rather than as an absolute measure. For a given occupation j or course c, Importance values are normalized *within each element type* (Skills, Knowledge, Tasks) to obtain weights $w_{j,x}^{(t)}$ that sum to one.

This design ensures that the model reflects how emphasis is distributed across elements within an occupation, rather than being sensitive to the absolute scale or reporting convention of O*NET Importance scores (e.g., 1–5 versus 0–100). Any transformation of Importance values for visualization or reporting is applied exclusively at the presentation layer and does not affect model computations.

3.5 Vectorization of Skills, Knowledge, and Tasks

To enable quantitative comparison between labor-market elements and AI impact, Skills, Knowledge, and Tasks are embedded into low-dimensional vector spaces that capture their dominant structural characteristics. Vectorization is performed separately for each element type using element-specific coordinate systems, as defined in Section 3.2.

- Skills in CTS space (Cognitive, Technical, Social),
- Knowledge in NHP space (Natural, Human, Formal),
- Tasks in NIP space (Natural/Physical, Information, People).

For each element x of type t, we define a raw vector

$$r_x^{(t)} \in [0, 1]^3$$

whose components represent the relative contribution of each dimension within the corresponding space. To ensure comparability across elements, vectors are subsequently normalized using the ℓ_1 norm:

$$v_x^{(t)} = \frac{r_x^{(t)}}{\|r_x^{(t)}\|_1}$$

3.5.1 Illustrative Examples and Vector Norm Construction

We provide one representative example for each element type to illustrate how vectors are constructed and normalized.

Skill Example: Critical Thinking Consider the skill *Critical Thinking*. This skill primarily involves reasoning and evaluation, with limited technical execution and minimal interpersonal interaction. We assign the raw CTS vector:

$$r_{\text{CT}}^{(\text{CTS})} = (0.75,\ 0.15,\ 0.10)$$

corresponding to Cognitive, Technical, and Social dimensions, respectively. The ℓ_1 norm of this vector is

$$\|r_{\text{CT}}^{(\text{CTS})}\|_1 = 0.75 + 0.15 + 0.10 = 1.00$$

so the normalized vector equals the raw vector:

$$v_{\text{CT}}^{(\text{CTS})} = (0.75,\ 0.15,\ 0.10)$$

This representation encodes the internal composition of the skill rather than its overall importance. Similar rules apply for elements in Knowledge and Tasks in later model computations.

Norm Determination and Interpretability Raw component values are determined using a structured rubric informed by O*NET task descriptions, domain definitions, and expert judgment. Importantly, the magnitude of the raw vector is not meaningful; only the *relative proportions across dimensions* are used. ℓ_1 normalization ensures that all vectors lie on a simplex, enabling consistent interpretation of dot products as alignment measures in subsequent sections.

This vectorization strategy guarantees that later index computations reflect *structural alignment* between AI impact and labor elements, rather than absolute scale or reporting artifacts.

3.5.2 Similarity and Distance Measures

Once Skills, Knowledge, and Tasks are embedded as normalized vectors in their respective element-specific spaces, standard geometric measures can be used to quantify relationships between elements of the same type.

Cosine Similarity. For two elements a and b represented by vectors $v(a)$ and $v(b)$ in the same space, cosine similarity is defined as

$$\mathrm{sim}(a, b) = \frac{v(a) \cdot v(b)}{\|v(a)\| \, \|v(b)\|}$$

Cosine similarity measures directional alignment and captures similarity in structural composition across dimensions.

Euclidean Distance. The Euclidean distance between two elements is defined as

$$d(a, b) = \|v(a) - v(b)\|$$

which measures absolute separation in the embedding space and is useful for clustering and neighborhood analysis.

Role in the Model. In this framework, cosine similarity and Euclidean distance are primarily used for exploratory analysis and visualization. Core indices introduced in later sections rely on dot products between aligned vectors, which—under ℓ_1 normalization—can be interpreted as composition-weighted alignment scores closely related to cosine similarity.

4 AI Impact Vectorization and AI Exposure Index

4.1 Motivation and Structural Alignment

To move from qualitative discussion of generative artificial intelligence (Gen-AI) impact to a quantitative, decision-oriented framework, we require a structured mechanism to evaluate how Gen-AI interacts with specific skills, knowledge areas, and tasks. Treating Gen-AI as a single, uniform technology acting equally across all domains is insufficient to analyze how AI reshapes work and learning.

Accordingly, this framework evaluates Gen-AI influence at the dimensional level, focusing on each of the three dimensions of skills, knowledge, and tasks as the fundamental units of analysis. Gen-AI impact is represented in the same vector spaces introduced in Section 3, allowing AI influence to be evaluated relative to the internal structure of each element. This enables precise, interpretable observation on AI.

4.2 AI Exposure Levels (A0–A4): Conceptual Definition

While element-level alignment scores provide a continuous measure of Gen-AI relevance, educational and policy decisions often require *categorical interpretations*. To support such decisions, we define five discrete **AI Exposure Levels**, denoted **A0** through **A4**, which characterize qualitatively distinct modes of interaction between Gen-AI and labor-market elements.

Each level corresponds to a different functional role of Gen-AI, ranging from minimal relevance to full substitutability:

- **A0 — AI Resistant**: Gen-AI has minimal relevance to the element. Core performance relies on human capability, and current Gen-AI systems neither assist nor meaningfully replace the element.
- **A1 — AI Assisted**: Gen-AI provides auxiliary support that improves efficiency or convenience without altering the fundamental structure of the element. Human expertise remains primary, with AI serving a supportive role.
- **A2 — AI Augmented**: Gen-AI enhances performance by expanding capability, speed, or scope. Human–AI collaboration becomes integral, but the element remains human-centered.
- **A3 — AI Compressed**: Gen-AI substantially reduces the time, effort, or skill required to perform the element. Traditional pathways for developing competence may be shortened or bypassed, and the element's standalone importance is diminished.
- **A4 — AI Substitutable**: Gen-AI can perform the element at a level comparable to or exceeding typical human performance in many contexts. Human involvement becomes optional, supervisory, or exceptional.

These five exposure levels are defined on the unit interval $[0, 1]$ and correspond to increasing degrees of Gen-AI alignment and substitutability.

4.3 Continuous AI score

Let $\tilde{I}_x \in [0, 1]$ denote the normalized AI score for a dimension of an element x (skill, knowledge area, or task), as defined in the previous section. This score captures the degree to which the *structural pattern* of Gen-AI impact aligns with the composition of the element.

The continuous alignment score serves as the quantitative backbone of the AI Exposure Index. Rather than relying on binary judgments of whether AI "affects" an element, the model positions each element along a continuous spectrum of AI exposure.

4.4 Mapping Continuous Scores to A0–A4 Exposure Levels

Each element is assigned to one of the five AI Exposure Levels by partitioning the unit interval into five bins:

$$A(x) = \begin{cases} \text{A0 (AI Resistant)} & 0 \le \tilde{I}_x < 0.2 \\ \text{A1 (AI Assisted)} & 0.2 \le \tilde{I}_x < 0.4 \\ \text{A2 (AI Augmented)} & 0.4 \le \tilde{I}_x < 0.6 \\ \text{A3 (AI Compressed)} & 0.6 \le \tilde{I}_x < 0.8 \\ \text{A4 (AI Substitutable)} & 0.8 \le \tilde{I}_x \le 1.0 \end{cases}$$

Threshold values shown here are illustrative and can be recalibrated using empirical data and institution norms. Importantly, while thresholds may change, the AI level remains stable.

4.5 Skill Index Construction and Alignment Interpretation

For skills, Gen-AI alignment is computed in the CTS space. Each skill $s \in \mathcal{S}$ is represented by a normalized structural vector

$$v_s^{(\text{CTS})} = \frac{r_s^{(\text{CTS})}}{\|r_s^{(\text{CTS})}\|_1}$$

and a corresponding normalized Gen-AI impact vector

$$a_s^{(\mathrm{CTS})} = \frac{q_s^{(\mathrm{CTS})}}{\|q_s^{(\mathrm{CTS})}\|_1}$$

The **Skill Alignment Index** is defined as the dot product:

$$I_s = \left\langle v_s^{(\mathrm{CTS})},\ a_s^{(\mathrm{CTS})} \right\rangle = \sum_{i=1}^{3} v_{s,i}^{(\mathrm{CTS})}\, a_{s,i}^{(\mathrm{CTS})}$$

This formulation measures the extent to which Gen-AI primarily affects the same dimensions that dominate the skill itself. Skills with higher alignment scores are more likely to fall into higher AI exposure categories (A2–A4).

4.6 Extension to Knowledge and Tasks

The same alignment-based logic applies to other element types using their respective vector spaces:

- **Knowledge elements** are evaluated in the NHP space:

$$I_k = \left\langle v_k^{(\mathrm{NHP})},\ a_k^{(\mathrm{NHP})} \right\rangle$$

- **Tasks** are evaluated in the NIP space:

$$I_t = \left\langle v_t^{(\mathrm{NIP})},\ a_t^{(\mathrm{NIP})} \right\rangle$$

For each element type, the resulting alignment score is normalized and mapped to an AI Exposure Level (A0–A4), enabling consistent interpretation across skills, knowledge, and tasks.

5 Occupation-Level AI Exposure Index (Job Index)

At the occupational level, Gen-AI exposure is summarized using a **Job Index**, which aggregates element-level AI indices across skills, knowledge areas, and tasks associated with a given occupation. Aggregation is weighted by occupation-specific importance measures, ensuring that elements central to job performance contribute proportionally more to the final index.

Let $j \in \mathcal{J}$ denote an occupation, and let x index elements (skills, knowledge, or tasks) associated with that occupation. For each element type $t \in \{\mathcal{S}, \mathcal{K}, \mathcal{T}\}$, we define normalized importance weights

$$w_{j,x}^{(t)} = \frac{\mathrm{Imp}_{j,x}}{\sum_{x' \in \mathcal{X}_j^{(t)}} \mathrm{Imp}_{j,x'}} \qquad \sum_x w_{j,x}^{(t)} = 1$$

where $\mathrm{Imp}_{j,x}$ denotes the occupation-specific importance score of element x.

Let $I_x^{(t)}$ denote the element-level AI alignment index defined in Section 4. The within-layer Job Index for occupation j and element type t is computed as a weighted sum:

$$J_j^{(t)} = \sum_{x \in \mathcal{X}_j^{(t)}} w_{j,x}^{(t)}\, I_x^{(t)}$$

This formulation represents a product–sum aggregation: each element's AI index is multiplied by its relative importance and summed across all elements of the same type.

To obtain a single occupation-level Job Index, we aggregate across element types using cross-layer weights:

$$J_j = \alpha_S J_j^{(\mathcal{S})} + \alpha_K J_j^{(\mathcal{K})} + \alpha_T J_j^{(\mathcal{T})}, \qquad \alpha_S + \alpha_K + \alpha_T = 1$$

The resulting Job Index $J_j \in [0, 1]$ summarizes the overall degree of Gen-AI exposure for occupation j. Higher values indicate that the core elements of the occupation are more structurally aligned with Gen-AI impact, suggesting greater potential for AI adoption, augmentation, or substitution.

6 Course Index with School-Specific Weighting Matrix

6.1 Course-Level Aggregation Framework

Courses differ systematically across post-secondary schools in how they emphasize skills, knowledge, and tasks. To account for this heterogeneity, we define a **Course Index** that aggregates element-level AI indices using school-specific weighting.

Let $c \in C$ denote a course. Through curriculum mapping, each course is associated with three sets of elements:

$$\mathcal{S}_c \subseteq \mathcal{S}, \qquad \mathcal{K}_c \subseteq \mathcal{K}, \qquad \mathcal{T}_c \subseteq \mathcal{T},$$

corresponding to skills, knowledge areas, and tasks emphasized in the course. Each element x has an element-level AI alignment index $I_x^{(t)}$ defined in Section 4.

6.2 Sector–Element Weighting Matrix

We introduce a **school-specific weighting matrix** $\tau_{s,x}$ to account for structural differences across post-secondary education.

Let the row index correspond to the educational sector

$$s \in \{\text{STEM}, \text{Trade}, \text{Arts}\}$$

and the column index correspond to the elements

$$x \in \{\mathcal{S}, \mathcal{K}, \mathcal{T}\}$$

The weighting matrix is defined as:

$$\tau = \begin{pmatrix} \tau_{\text{STEM},\mathcal{S}} & \tau_{\text{STEM},\mathcal{K}} & \tau_{\text{STEM},\mathcal{T}} \\ \tau_{\text{Trade},\mathcal{S}} & \tau_{\text{Trade},\mathcal{K}} & \tau_{\text{Trade},\mathcal{T}} \\ \tau_{\text{Arts},\mathcal{S}} & \tau_{\text{Arts},\mathcal{K}} & \tau_{\text{Arts},\mathcal{T}} \end{pmatrix} \qquad \tau_{s,x} \in [0, 1]$$

Each row of τ is normalized such that

$$\tau_{s,\mathcal{S}} + \tau_{s,\mathcal{K}} + \tau_{s,\mathcal{T}} = 1 \qquad \forall i$$

6.3 Within-Type Course Indices

Similar to the job index, for each course c and element type $t \in \{\mathcal{S}, \mathcal{K}, \mathcal{T}\}$, we define normalized course coverage weights $w_{c,x}^{(t)}$ satisfying

$$\sum_{x \in \mathcal{X}_c^{(t)}} w_{c,x}^{(t)} = 1$$

where

$$\mathcal{X}_c^{(\mathcal{S})} = \mathcal{S}_c, \quad \mathcal{X}_c^{(\mathcal{K})} = \mathcal{K}_c \quad \mathcal{X}_c^{(\mathcal{T})} = \mathcal{T}_c$$

The within-type course indices are computed as

$$C_c^{(t)} = \sum_{x \in \mathcal{X}_c^{(t)}} w_{c,x}^{(t)} I_x^{(t)}$$

6.4 Sector-Adjusted Course Index

Given that course c belongs to sector i, the overall **Course Index** is defined as

$$C_c^{(i)} = \tau_{s,\mathcal{S}} C_c^{(\mathcal{S})} + \tau_{s,\mathcal{K}} C_c^{(\mathcal{K})} + \tau_{s,\mathcal{T}} C_c^{(\mathcal{T})}$$

The resulting index $C_c^{(i)} \in [0, 1]$ summarizes the degree of Gen-AI exposure embedded in course c, adjusted for school-specific curriculums.

7 STEM Career and Institution Case

7.1 Actuaries

Actuaries analyze statistical data (e.g., mortality, accident, sickness, disability, retirement rates) to forecast risk and future benefit liabilities, then translate that analysis into pricing and financial decisions for insurers, pension funds, and related organizations. Typical work includes estimating event rates from data, setting premium rates and required cash reserves, designing/reviewing insurance or pension plans for financial soundness, and communicating technical findings to executives, regulators, or other stakeholders.

7.1.1 Decomposition

To model the occupation "Actuaries", we decompose it into the same computable elements used throughout the framework: Skills, Knowledge, and Tasks (See Sections 3.2&3.5.1). For every element associated with Actuaries, we attach (i) an AI influence vector defined in the same 3D space so it can interact directly with the element vector (See Section 4.2), and (ii) an occupation-specific importance score capturing how central that element is to actuarial work (See Section 3.4).

This produces a case-specific input bundle: a structured table of Actuaries' vector-space representations, their matched AI vectors, and their importance scores.

7.1.2 Model Implementation

Using the element bundle defined above, we then compute element-level penetration indices and an occupation-level summary index for Actuaries. As explained in the modeling section earlier, each Skill, Knowledge, and For each element, we first calculate a raw AI–element interaction score and then convert

it into a standardized element index so that indices are comparable across all Skills, Knowledge items, and Tasks within the occupation.

$$\tilde{I}_x^{(t)} = r_x^{(t)} \cdot a_x^{(t)} \qquad t \in \{\mathrm{S}, \mathrm{K}, \mathrm{T}\} \tag{1}$$

$$I_x^{(t)} = \frac{\tilde{I}_x^{(t)} - \min_{y \in t} \tilde{I}_y^{(t)}}{\max_{y \in t} \tilde{I}_y^{(t)} - \min_{y \in t} \tilde{I}_y^{(t)}} \tag{2}$$

We then combine these element indices using the Actuaries-specific importance scores from the inputs, producing an aggregated value, occupation-level AI exposure index, or job index (See Section 5), while preserving full traceability back to the element layer (i.e., every contribution to the aggregate can be decomposed into the underlying element index and its importance weight).

$$J_j = \sum_{t \in \{\mathrm{S,K,T}\}} \alpha_t \sum_{x \in t} w_{j,x}^{(t)} I_x^{(t)} = joe \tag{3}$$

7.1.3 Interpretation

The Occupation-Level AI Exposure Index (the Job Index) should be interpreted as a content-based measure of how strongly an occupation's underlying skill, knowledge, and task profile aligns with the capability footprint of AI. It is not a labor-market forecast; rather, it is a structural score derived from representing the job as a collection of element vectors and then quantifying AI's influence on those elements through vector alignment, so the index estimates how much of the job's "work content" sits in regions where AI can meaningfully automate or augment.

Against a broader background, a higher Job Index $J_j \in [0, 1]$ means that the occupation's most important skills, knowledge areas, and working activities have greater overlap with the AI-impact vectors, implying stronger potential for AI-driven transformation of how the work is executed. This does not automatically mean displacement; it often indicates that AI will become a core productivity layer and that the occupation's human value may shift toward oversight, judgment, integration, and higher-level decision-making.

Specifically, the occupation "Actuaries" here has an Occupation-Level AI Exposure Index (Job Index) of [JOE], which places it at the [Joe]th percentile in the overall distribution constructed from 800+ O*NET occupations. This positioning suggests that the work content of actuaries—when represented through its underlying skills, knowledge areas, and tasks—shows a relatively strong alignment with the model's AI-impact vectors compared with most occupations, implying that AI is likely to meaningfully reshape how actuarial work is performed (through automation and augmentation of key analytical and information-processing components) rather than leaving the occupational task mix largely unchanged.

Lastly, because the job index is built from assumed AI-impact vectors, its meaning is conditional on those assumptions and is most responsibly communicated in relative terms, such as normalized scores or percentiles within the occupation set. In practice, the most actionable interpretation comes from decomposing the Job Index into its skill, knowledge, and task contributors, since that diagnosis explains which specific elements are driving exposure and therefore where education or training should incorporate AI most directly.

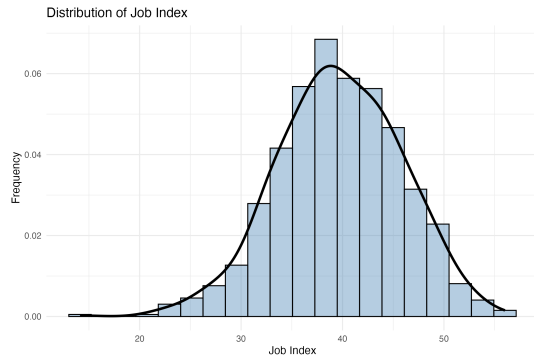

Figure 1

7.2 Wake Forest University – B.S. Statistics Program

As a private research university in Winston-Salem, Wake Forest's STEM-facing value is less about "training for one job" and more about turning broad quantitative and research training into first-destination outcomes: full-time roles plus graduate/professional school.

7.2.1 Decomposition

Here, we evaluate the course "Statistical Learning" at Wake Forest University within our framework by treating the course as a structured bundle of elements and decomposing it. As explained in the modeling section earlier, each decomposed element is thenrepresented as a 3-dimensional vector in its corresponding space so that element-level AI exposure can be computed consistently and compared across courses (See Sections 3.2 & 3.5.1). For this course-specific application, we then attach two additional pieces of information to each element: (i) an AI impact vector and (ii) a course-assigned importance (See Sections 4.2 & 3.4).

Finally, because this is a STEM course, we specify top-level aggregation weights to reflect the expected distribution of instructional emphasis across element families using the weighting matrix.

$$\tau_{\text{STEM},x} = \left[\tau_{\text{STEM},\mathcal{S}},\ \tau_{\text{STEM},\mathcal{K}},\ \tau_{\text{STEM},\mathcal{T}}\right] = \left[0.3,\ 0.5,\ 0.2\right]$$

which serve as the course-level weighting scheme for synthesizing element-level results into interpretable course outputs (See Section 6.2).

Table 3: Selection of Skills for Statistical Learning

Skill Name	Vector	AI Index	Importance Score
Mathematics	(0.8, 0.9, 0.05)	(0.75, 1, 1)	0.333
System Analysis	(0.9, 0.75, 0.3)	(1, 0.75, 0.75)	0.333
Programming	(0.8, 0.95, 0.1)	(0.75, 1, 1)	0.333

Table 4: Selection of Knowledge Areas for Statistical Learning

Knowledge Area	Vector	AI Index	Importance Score
Telecommunications	(0.75, 0.7, 0.3)	(0.75, 0.75, 0.75)	0.333
Computers and Electronics	(0.9, 0.65, 0.15)	(1, 0.75, 0.75)	0.333
Engineering and Technology	(0.6, 0.85, 0.25)	(0.5, 1, 0.75)	0.333

Table 5: Selection of Tasks for Statistical Learning

Task	Vector	AI Index	Importance Score
Analyzing Data or Information	(0.95, 0.1, 0.05)	(1, 1, 1)	1
Developing Objectives and Strategies	(0.9, 0.35, 0.05)	(1, 0.75, 1)	1
Working with Computers	(0.9, 0.1, 0.1)	(1, 1, 1)	1

7.2.2 Model Implementation

Using the element set constructed for Statistical Learning, we compute element-level indices by interacting each element's 3-dimensional representation with its corresponding 3-dimensional AI impact representation, as defined earlier. Concretely, for every Skill in the CTS space, every Knowledge element in the MIP space, and every Task in the NHF space, we apply the same core operation described in the modeling section to produce a raw element score, then place these raw scores onto a comparable scale via the normalization procedure specified earlier. This yields a complete set of Skill Indices, Knowledge Indices, and Task Indices that quantify AI penetration at the element level for the course (See Section 4.5).

$$C_c^{(t)} = \sum_{x \in \mathcal{X}_c^{(t)}} w_{c,x}^{(t)} \, I_x^{(t)}$$

joe

By recalling the model component defined in 6.4, we synthesize an overall Course Index by combining the three family sub-indices using the STEM weighting scheme so the final course measure reflects both (i) which elements are AI-penetrated and (ii) how strongly the course emphasizes them.

$$C_c^{(i)} = \tau_{s,\mathcal{S}} \, C_c^{(\mathcal{S})} + \tau_{s,\mathcal{K}} \, C_c^{(\mathcal{K})} + \tau_{s,\mathcal{T}} \, C_c^{(\mathcal{T})} \; = joe$$

7.2.3 Interpretation

The Course Index of [Joe] for Wake Forest University's Statistics B.A. is interpreted not as a relative "rank" on an exposure curve, but as a practical signal of instructional urgency: a higher Course Index means that incorporating GenAI-related capabilities is more necessary for students to achieve the course's learning objectives under current (and near-future) work conditions. Put differently, the Course Index is an "AI-integration pressure" score for the class itself—indicating how strongly the subject matter, typical student outputs (problem sets, projects, code, reports), and downstream professional expectations are likely to intersect with GenAI-augmented workflows.

Once the Course Index indicates non-trivial urgency, the actionable step is to open the black box and use the element indices—Skill/Knowledge/Task indices—together with the School-Specific Weighting

Matrix to identify where the program's AI pressure is actually coming from and which components of the curriculum are carrying that pressure. For example, here, for Wake Forest's Statistics Program

SkillName Index - importance[Joe] KnowledgeName Index - importance[Joe] TaskName Index - importance[Joe]

Concretely, you would (i) locate the skills/knowledge/tasks that jointly have high exposure indices and high weights (importance × school-specific weights), (ii) audit how explicitly those elements are embedded in syllabi and assessments (what is taught, what is practiced, what is graded), and (iii) translate any gaps into targeted curriculum decisions rather than generic "add an AI week." This is also the point where the program becomes situational: you validate whether faculty already plan to incorporate GenAI, and you design integration that matches the course's work products—e.g., building assignments that require students to critique AI-assisted analysis, document model choices and assumptions, stress-test outputs, and demonstrate statistical reasoning and communication that remains defensible under AI assistance—so GenAI is treated as an accountability-bound tool within statistics training, not a parallel topic bolted on.

8 Trade Career and Institution Case

8.1 HVAC Technicians

HVAC technicians install, maintain, and repair climate-control and refrigeration systems to keep indoor environments comfortable, efficient, and safe in homes, commercial buildings, and industrial facilities. Core tasks include installing or replacing equipment and ductwork or piping, testing system performance, diagnosing malfunctions, and performing repairs such as fixing leaks, replacing components, and recharging or recovering refrigerants.

8.1.1 Model Implementation

To model HVAC Technicians, we again decompose the occupation into our three-element families. To avoid repetitiveness, we will not re-present the underlying dataframes or individual element-level values here; instead, we proceed directly to computing the occupation's aggregate Job Index using the already-defined element vectors, importance scores, and AI impact vectors (For computational process, see Section 7.1).

$$J_j = \sum_{t \in \{S,K,T\}} \alpha_t \sum_{x \in t} w_{j,x}^{(t)} I_x^{(t)} = joe \tag{4}$$

8.1.2 Interpretation

That job index of [JOE] places the occupation at roughly the [X]th percentile on the AI-exposure curve built from 800+ O*NET occupations.

Given this output, stakeholders may want to dive deeper into the element indices which indicate where AI pressure is most likely to accumulate within the job's work content. Specifically, [joe] skill has a high importance as [joe] for HVAC technicians, showing HVAC work is constrained by on-site conditions, equipment variability, safety codes, and physical execution, i.e., end-to-end automation is limited. While AI is more likely to augment the information-intensive portions of the workflow—such as diagnostics support, documentation, quoting, scheduling, and compliance-oriented checks—than to substitute core hands-on troubleshooting and repair.

8.2 Forsyth Tech – Air Conditioning, Heating, and Refrigeration Technology Program

Forsyth Technical Community College (Forsyth Tech) is a community college serving the local region and offering workforce-oriented programs (including skilled trades pathways). It was founded in 1960 and positions itself around workforce readiness and "the future of work," making it a natural fit for a trade-school example where curriculum-to-occupation alignment can be evaluated at the course/module level.

8.2.1 Model Implementation

To model the Air Conditioning, Heating, and Refrigeration Technology Program, we apply the same element-level decomposition. However, as this is a trade-school program, we aggregate the three-element families using an explicitly task-forward weighting scheme

$$\tau_{\text{Trade},\,x} = \left[\tau_{\text{Trade},\mathcal{S}},\ \tau_{\text{Trade},\mathcal{K}},\ \tau_{\text{Trade},\mathcal{T}}\right] = \left[0.3,\ 0.2,\ 0.5\right]$$

to reflect the program's practice- and execution-oriented learning design. To avoid repetitiveness, we do not reproduce the underlying data tables or individual element values here; instead, we proceed directly to the weighted index computation and the resulting program-level AI penetration assessment (For computational process, see Section 7.2).

$$C_c^{(i)} = \tau_{s,\mathcal{S}}\, C_c^{(\mathcal{S})} + \tau_{s,\mathcal{K}}\, C_c^{(\mathcal{K})} + \tau_{s,\mathcal{T}}\, C_c^{(\mathcal{T})} \ = joe$$

8.2.2 Interpretation

The Course Index of [JOE] for the Air Conditioning, Heating, and Refrigeration Technology program is a practical cue for how much GenAI should be integrated into teaching and assessment.

If [JOE] is meaningfully high, the next step is to trace where that pressure comes from by pairing the element indices with the School-Specific Weighting Matrix and surfacing the main drivers as [Element-Name — Index — Importance — SchoolWeight — Contribution] for skills, knowledge, and tasks. You then check whether those drivers are already trained and graded in the curriculum, and convert any gaps into specific integration points—typically around diagnostics reasoning, documentation and compliance, estimating/quoting, and customer communication—rather than adding generic "AI content."

9 Arts Career and Institution Case

9.1 Dancers

Dancers express ideas and stories through movement by rehearsing and performing choreographed or improvised routines for live audiences, film, or other productions. Typical tasks include rehearsing dance routines, auditioning for roles, performing in front of audiences or cameras, and coordinating timing and spacing with other performers; many also learn new choreography quickly and maintain physical conditioning as part of ongoing professional practice.

9.1.1 Model Implementation

To model Dancers, we decompose the occupation into Skills, Knowledge, and Tasks. To avoid repetition, we omit element-level inputs here and proceed directly to the index computation using the already-defined vectors, importance scores, and AI impact vectors (For computational process, see Section 7.1).

$$J_j = \sum_{t \in \{\mathrm{S,K,T}\}} \alpha_t \sum_{x \in t} w_{j,x}^{(t)} I_x^{(t)} = joe \tag{5}$$

9.1.2 Interpretation

Dancers' Job Index is [JOE], placing the occupation at roughly the [X]th percentile on the AI-exposure curve built from 800+ O*NET occupations.

Given this output, stakeholders may want to look beneath the occupation-level score and examine the element indices to see where AI pressure is most likely to accumulate within dancers' work content. Specifically, the [JOE] skill carries a high importance of [JOE] for dancers, indicating that the core of professional dance remains constrained by embodied technique, real-time responsiveness, and performance-based evaluation in live or production settings, which makes end-to-end automation structurally limited. In practice, AI is more likely to augment adjacent, information- and media-intensive parts of the workflow—such as choreography ideation support, music editing and mixing, rehearsal planning and annotation, video review and feedback, audition materials and portfolio packaging, and marketing/distribution—than to substitute the central human elements of physical execution, interpretive artistry, ensemble coordination, and stage presence.

9.2 UNC School of the Arts – B.A. Contemporary Dance Program

UNCSA's value to arts careers is that it functions like a professional production pipeline: students train in conservatory-style programs that map directly onto real-world creative labor markets (stage, film/TV, production crafts). Importantly, this "where graduates go" logic is not limited to performers—UNCSA highlights backstage and production-specialist trajectories (e.g., hair/makeup, sound, design, and production) that feed into the same professional ecosystems as on-stage talent.

9.2.1 Model Implementation

To model the UNC School of the Arts – B.A. Contemporary Dance Program, we again represent the program's curriculum as a structured combination of Skills, Knowledge, and Tasks. Because this is an arts school program, we apply program-level weights of

$$\tau_{\mathrm{Art},x} = \left[\tau_{\mathrm{Art},\mathcal{S}},\ \tau_{\mathrm{Art},\mathcal{K}},\ \tau_{\mathrm{Art},\mathcal{T}}\right] = \left[0.5,\ 0.2,\ 0.3\right]$$

in order to adjust for the importance of elements (For computational process, see Section 7.2).

$$C_c^{(i)} = \tau_{s,\mathcal{S}}\, C_c^{(\mathcal{S})} + \tau_{s,\mathcal{K}}\, C_c^{(\mathcal{K})} + \tau_{s,\mathcal{T}}\, C_c^{(\mathcal{T})} = joe$$

9.2.2 Interpretation

UNC School of the Arts' B.A. Contemporary Dance program has a Course Index of [JOE], which we treat as a direct indicator of how much GenAI literacy should be embedded into the program's core training and evaluation.

When [JOE] is non-trivial, the next move is to pinpoint where that pressure actually comes from: apply the School-Specific Weighting Matrix to the Skill/Knowledge/Task indices, then surface the top contributors in the format [ElementName — Index — Importance — SchoolWeight — Contribution]. Those drivers are then checked against what students repeatedly do and get judged on—technique and conditioning routines, partnering and rehearsal discipline, repertory learning, composition and improvisation processes, reflection and research, and career-facing deliverables—and any mismatches become precise integration decisions (e.g., AI-assisted research/context building, structured critique and iteration logs, portfolio/reel and written materials, scheduling/production coordination), keeping the artistic decision-making and embodied execution as the non-delegable center.

10 Model Assessment and Critiques

10.1 Scalability

In the current prototype, we represent Skills as vectors in a 3D CTS space—Cognitive, Technical, Social—primarily to support intuitive visualization and explainability (e.g., a skill becomes a point in a coordinate system, and its vector form is easy to interpret and compare). However, this 3D choice is a modeling convenience rather than a structural constraint. In real deployment, the same vectorization logic can be generalized to higher-dimensional embedding spaces (e.g., adding dimensions such as creativity, physicality, regulatory risk, teamwork intensity, uncertainty tolerance, etc.), which can improve descriptive accuracy and reduce information loss from coarse categorization. The same scalability applies symmetrically to Knowledge and Tasks/Work Activities: each element can be mapped into an n-dimensional space, and downstream computations (e.g., similarity, dot products, weighted aggregation into course/job indices) remain mathematically consistent.

Similarly, we currently define a single AI Index as a 3D vector that approximates how "AI influence" projects onto the CTS structure of skills, knowledge, and tasks. But as AI systems diversify and the underlying foundation-model landscape evolves, we can extend this into a family of AI indices rather than a single universal one. For instance, separate indices can be constructed for language-model impact, vision-model impact, audio-model impact, and even physical/robotics-model impact, reflecting that different model modalities substitute, augment, or transform different parts of human work in fundamentally different ways. Because our framework treats "AI impact" as an input vector (or a set of vectors), it naturally supports multiple indices, higher-dimensional indices, and modular updates over time—allowing the model to remain stable while the AI ecosystem changes. In other words, while "GenAI" is still not a fixed category, our architecture is designed to accept richer, more granular AI impact representations without rewriting the pipeline.

10.2 Reliability and Synthetic Data Issues

Secondly, the current vectorization pipeline uses custom scraping and generative-AI-assisted mapping to assign elements into their vector spaces, which introduces a degree of synthetic inference at the representation stage. This does not undermine the model's reliability as an architecture, because the model is explicitly designed so that these coordinates can be replaced by empirical measurements whenever available.

Our team very well acknowledges the built-in complexity of STEM, trade, and arts programs and how their curriculum may easily overlap with each other. Hence, in practical use, we can upgrade the coordinates or even the construction of the vector space of any element through survey-based or other empirical data

collection, improving measurement validity and allowing calibration across industries, schools, and learner populations. For example, if we currently place "Critical Thinking" at

$$r_{\mathrm{CT}}^{(\mathrm{CTS})} = (0.75,\ 0.15,\ 0.10)$$

in the CTS space, we can design targeted instruments to ask both industry practitioners and students across programs of study how they perceive the skill's cognitive/technical/social composition. A sample survey item could be:

On a scale of 1–7, how "cognitive" do you think the skill "Critical Thinking" is?

Similarly, while working with educational institutions, we can have administrators, lecturers, and other related personnel to help determine the importance of skill, knowledge, and task elements of a course, as well as how each family of elements is weighted under different types of post-secondary educational institutions. This empirical layer makes the model generalizable across contexts (industry, region, institution type) and supports iterative refinement: as more data accumulates, coordinate estimates become more stable, more defensible, and more aligned with real-world perceptions and outcomes.

10.3 Generalizability

Though indexes are created in the model-building process, those indexes themselves are deemed useful in assessing personal growth, self-assessment, and career potential.

10.4 Auditability and Reproducibility

11 Conclusion